

WAHOO, NE, USA

WMO: 720942

Lat: 41.241N

Lon: 96.594W

Elev: 373

StdP: 96.92

Time Zone: -6.00 (NAC)

Period: 11-19

WBAN: 00323

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.0	-16.3	-23.8	0.5	-17.5	-21.3	0.6	-15.5	15.7	-2.1	14.0	-1.4	3.1	330	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	35.2	22.8	33.3	22.9	31.9	22.8	25.9	31.2	25.0	30.3	24.1	29.4	5.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.6	20.5	29.2	23.4	19.0	28.0	22.6	18.1	27.2	82.6	31.2	78.2	30.4	74.5	29.4	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	10.9	9.3	-23.8	37.4	2.3	1.6	-25.4	38.6	-26.8	39.6	-28.1	40.5	-29.7	41.7		
(5)			-24.1	27.9	2.3	0.9	-25.7	28.5	-27.0	29.0	-28.3	29.5	-29.9	30.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	-3.6	-3.0	5.2	10.7	17.5	23.4	24.7	22.6	19.6	11.9	4.0	-1.6		
	DBStd	11.46	6.34	6.85	7.05	5.50	5.13	3.45	3.23	2.83	4.57	4.96	5.74	6.07		
	HDD10.0	1617	421	364	181	57	6	0	0	0	1	38	188	360		
	HDD18.3	3297	679	597	409	232	79	3	1	2	39	207	429	618		
	CDD10.0	1993	0	1	32	79	238	401	457	390	289	95	9	1		
	CDD18.3	631	0	0	2	4	53	154	200	134	78	6	0	0		
	CDH23.3	6385	0	0	44	93	590	1591	2074	1121	775	93	4	0		
	CDH26.7	2588	0	0	11	26	229	676	924	409	296	17	0	0		
(14) Wind	WSAvg	4.1	4.4	4.6	4.9	5.4	4.6	4.2	2.9	2.7	3.3	4.0	4.4	4.2		
(15) Precipitation	PrecAvg	766	19	23	41	76	119	125	89	99	69	52	35	23		
	PrecMax	1025	42	63	88	129	218	275	269	292	181	143	75	61		
	PrecMin	514	4	3	0	13	34	33	18	12	16	3	1	0		
	PrecStd	134	10	14	26	26	52	59	55	67	40	36	22	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.1	20.6	27.7	30.1	35.6	36.4	37.4	36.2	34.7	28.7	23.2	15.6		
		MCWB	7.3	10.7	16.0	16.6	18.3	22.6	23.6	22.6	21.7	16.8	13.5	8.4		
	2%	DB	10.6	13.7	23.8	25.7	31.9	34.3	35.1	32.9	32.3	26.0	18.8	12.2		
		MCWB	5.4	8.3	12.8	14.5	19.6	22.5	24.4	23.4	22.4	16.1	11.8	8.1		
	5%	DB	7.5	10.2	20.1	22.9	29.0	32.4	33.2	30.6	30.2	23.5	15.8	9.3		
		MCWB	3.3	5.9	12.0	13.4	18.0	22.1	24.5	23.5	21.9	14.6	9.5	5.4		
	10%	DB	5.0	6.2	16.6	20.6	26.7	30.6	31.6	28.8	27.8	20.8	13.0	6.5		
		MCWB	2.2	3.5	10.7	12.5	17.5	21.4	24.3	22.7	20.3	13.8	8.2	3.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	12.1	17.6	18.1	22.5	26.1	27.4	26.7	25.0	20.1	15.6	11.5		
		MCDB	13.9	15.7	24.8	25.2	28.7	32.2	32.8	31.4	30.5	24.3	19.9	13.1		
	2%	WB	5.5	8.3	15.4	16.2	20.8	24.8	26.4	25.2	23.9	18.0	12.5	7.8		
		MCDB	10.0	13.7	20.5	23.0	26.9	30.6	31.8	29.5	29.0	22.9	17.9	11.5		
	5%	WB	3.9	5.8	13.0	15.1	19.7	23.8	25.5	24.2	23.0	16.4	10.4	5.6		
		MCDB	7.2	9.8	19.5	20.9	26.8	29.7	31.0	28.8	28.0	20.7	15.1	8.9		
	10%	WB	2.5	3.6	10.7	13.8	18.7	22.8	24.6	23.2	22.1	14.7	8.3	3.7		
		MCDB	4.8	6.3	17.8	19.2	25.6	28.7	30.0	27.5	26.3	19.2	13.3	6.4		
(35) Mean Daily Temperature Range	MDBR		10.7	10.6	12.8	12.9	12.1	11.5	11.5	11.2	13.2	13.1	12.4	9.9		
	5% DB	MCDBR	15.0	15.9	19.7	17.9	15.8	13.5	13.2	14.1	15.1	17.2	17.1	14.7		
		MCWBR	10.3	10.3	10.4	8.8	5.6	5.1	5.6	6.3	6.2	8.6	10.1	10.1		
	5% WB	MCDBR	13.7	15.1	16.0	14.3	13.1	11.7	11.5	11.8	12.4	13.7	15.7	13.5		
		MCWBR	9.6	10.2	9.2	8.0	5.7	6.0	5.8	6.2	6.1	8.1	10.0	9.8		
(40) Clear-Sky Solar Irradiance	taub		0.281	0.309	0.330	0.386	0.409	0.418	0.425	0.417	0.378	0.331	0.295	0.282		
	taud		2.455	2.402	2.394	2.251	2.261	2.296	2.276	2.298	2.361	2.483	2.505	2.462		
	Ebn at Noon		883	903	920	884	869	859	848	844	856	867	859	849		
	Edh at Noon		76	94	106	132	134	130	131	125	109	86	72	70		
(44) All-Sky Solar Radiation	RadAvg		2.04	2.97	3.85	4.95	5.83	6.52	6.54	5.64	4.63	3.27	2.29	1.72		
	RadStd		0.19	0.17	0.34	0.34	0.43	0.47	0.41	0.36	0.30	0.37	0.15	0.17		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page